

N100 Financial Intelligence Platform - Final Report

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Project: N100 Financial Intelligence Platform

1. Project Overview

This project implements an end-to-end ETL and analytics pipeline for Nifty 100 financial datasets using Python, Pandas, SQLite, Pytest, Git and GitHub.

2. Objectives

- Load Excel datasets
- Validate and normalize data
- Store data in SQLite
- Compute financial KPIs
- Generate outputs
- Test the project

3. Datasets

Processed 12 datasets: analysis, balancesheet, cashflow, companies, documents, profitandloss, prosandcons, financial_ratios, market_cap, peer_groups, sectors and stock_prices.

4. Technologies

Python 3.11, Pandas, SQLite3, VS Code, Pytest, Git, GitHub.

5. ETL Pipeline

Loaded Excel files, cleaned column names, validated records and inserted data into SQLite.

6. Database

Created 12 SQLite tables. Verified financial_ratios contains 1184 rows.

7. Analytics

Implemented profitability, liquidity, efficiency, leverage, CAGR and cash flow KPI calculations.

8. Outputs

Generated CSV outputs including capital_allocation.csv and logs such as ratio_edge_cases.log.

9. Testing

Executed pytest successfully with all implemented tests passing.

10. GitHub

Project initialized, committed and pushed to GitHub.

11. Conclusion

The project successfully demonstrates a complete financial data Analytics workflow from ingestion to analytics, testing and reporting.